

ANANTH SURAMPUDI

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PROFILE

Full-Stack & GenAI Backend Engineer with 2+ years of experience building scalable web applications and real-time distributed systems using Python (FastAPI/Django), React/Next.js, PostgreSQL, and Redis.

Optimized p95 API performance under load, implemented reliable transactional checkout workflows, and integrated LLM-powered retrieval (RAG) systems into production environments, with a strong focus on concurrency control, scalable architecture, and end-to-end feature ownership.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Backend: FastAPI, Django, Node.js, REST APIs, Async/Await, Microservices, Scalable System Design

Databases: PostgreSQL, MySQL, MongoDB

Caching & Realtime: Redis (Cache-Aside, Pub/Sub), WebSockets

Frontend: React, Next.js, React Native, TailWind CSS, shadcn/ui, Zustand

AI & LLMs: Retrieval-Augmented Generation (RAG), vector search, FAISS, OpenAI APIs, semantic search

Cloud & Deployment: AWS, Docker, GitHub Actions

Testing: Jest, React Testing Library, Cypress, Playwright, Pytest, Unit & Integration Testing

WORK EXPERIENCE

Software Engineer

VWorker Solutions | Aug 2021 – Jul 2023

- Improved p95 API response times by ~30% by eliminating blocking I/O operations and optimizing high-cost PostgreSQL queries impacting user-facing workflows.
- Increased transactional throughput by ~25% through execution plan analysis and composite indexing strategies that reduced lock contention in high-traffic endpoints.
- Implemented Redis cache-aside patterns with structured invalidation logic, reducing database read pressure by ~40% during peak traffic.
- Strengthened checkout and order workflows by enforcing server-side validations to prevent stale pricing, inconsistent cart states, and race conditions during concurrent sessions.
- Implemented role-based access control across backend services and UI layers to enforce secure multi-user workflows.
- Expanded automated test coverage to 90%+, adding concurrency and payment retry scenarios to reduce regression defects in production releases.
- Collaborated with frontend, QA, and DevOps teams to validate migrations and deployment changes, ensuring smooth production rollouts with minimal downtime.

PROJECTS

Scalable E-Commerce Platform

Tech Stack: FastAPI, PostgreSQL, Redis, Elasticsearch, AWS

- Developed a high-traffic e-commerce application with reliable checkout, inventory management, and payment processing workflows designed to handle concurrent user activity.
- Implemented atomic inventory updates and strict server-side validations to prevent overselling, zero-item orders, stale pricing submissions, and tampered cart totals before payment authorization.
- Reduced p95 latency by ~30% during load testing (10k+ concurrent users) through async request handling, query optimization, and targeted indexing improvements.

- Introduced Redis caching for high-demand catalog and cart endpoints, lowering database read pressure by ~40% and stabilizing response times during peak traffic.
- Implemented retry-safe and idempotent payment logic to prevent duplicate charges during network failures or repeated client requests.
- Built hybrid product search combining lexical ranking and semantic retrieval to improve relevance for complex and natural-language queries.
- Deployed Dockerized backend services to AWS EC2, configuring production environment variables and reverse proxy routing to ensure reliable application availability.

Real-Time Ride-Hailing & Logistics Platform

Tech Stack: React, FastAPI, WebSockets, PostgreSQL, Redis, Stripe, AWS

- Developed a real-time ride tracking system enabling live driver location streaming and synchronized ride status transitions across distributed backend services and mobile clients.
- Managed 1k+ concurrent WebSocket connections using asynchronous processing, handling reconnects and stale sessions to prevent inconsistent driver visibility.
- Reduced end-to-end location update latency by ~40% by optimizing broadcast logic and leveraging Redis Pub/Sub for non-blocking message distribution across horizontally scaled instances.
- Implemented server-side ride state validation to prevent duplicate ride creation and enforce consistent status transitions across ride lifecycles.
- Integrated idempotent and retry-safe payment workflows to prevent duplicate charges during network instability.
- Collaborated with frontend engineers to define strict message contracts and validate real-time edge cases, preventing client-server desynchronization during rapid updates and connection drops.

Multi-Tenant Grocery Marketplace with AI Assistant

Tech Stack: React, FastAPI, PostgreSQL, Redis, OpenAI, AWS

- Developed a multi-tenant SaaS marketplace ensuring strict vendor data isolation within shared infrastructure, enforcing tenant-scoped queries and secure data boundaries across inventory and orders.
- Built responsive React-based web interfaces using modular component architecture and centralized state management to support checkout flows, cart updates, and real-time order tracking.
- Implemented role-based UI rendering and protected routes to dynamically control features and actions for customers, vendors, and administrators based on permission levels.
- Designed transaction-safe checkout workflows with server-side validation and synchronized inventory updates to prevent inconsistent order states during concurrent usage.
- Integrated an LLM-powered assistant using retrieval-augmented generation (RAG) with vector-based retrieval over live inventory and order data to improve answer accuracy.
- Coordinated frontend and backend validation layers, combining controlled form handling with server-side verification to prevent invalid submissions and maintain data integrity.

EDUCATION

Master of Science in Computer Science

Auburn University at Montgomery | Montgomery, AL